

Tejaswini Medi

Ph.D. Candidate in Machine Learning • Generative Models & Computer Vision

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Profile

Ph.D. candidate (expected completion early 2027) in the Machine Learning Group at the University of Mannheim, specialising in **high-fidelity generative modelling for 2D images and 3D shapes** and in **adversarial robustness and fairness** of deep neural networks. First author of papers at **ECCV, BMVC, WACV, GCPR** and an **ICCV workshop (oral)**. Industry research experience at **Autodesk Research** and **Endress+Hauser**, both resulting in peer-reviewed publications. Seeking Research Scientist / Applied Scientist roles in generative AI and computer vision.

Research Focus

Generative Modelling	Autoregressive & transformer-based generation, wavelet / frequency-aware image and 3D shape synthesis, implicit 3D representations, flow matching
Trustworthy ML	Adversarial training, class-wise robustness, fairness under adversarial attacks, RL for training optimisation
Representation Learning	Hierarchical (coarse-to-fine) representations for 2D and 3D data

Research Experience

Doctoral Researcher

Machine Learning Group, University of Mannheim

2023 – Present

Mannheim, Germany

- Proposed a frequency-decoupled image generation approach that separates high- and low-frequency content for faithful fine-detail synthesis (**ECCV 2026**).
- Designed targeted and mixed adversarial training methods (FAIR-TAT, TRIX) and an RL-based method (RL-FAT) that improve class-wise fairness of robust models (**WACV 2025, BMVC 2026**).
- Built reproducible PyTorch / PyTorch Lightning codebases with distributed training (Ray) on multi-GPU HPC clusters.

Research Intern

Autodesk Research, AI Lab

Jul 2024 – Sep 2024

- Developed **3D-WAG**, a hierarchical autoregressive 3D generative model built on wavelet-based representations that captures coarse-to-fine shape detail.
- Achieved fast inference while producing high-fidelity 3D shapes; work published at **BMVC 2025**.
- Bridged academic research with industrial 3D design workflows in collaboration with Autodesk scientists.

Student Researcher

Endress+Hauser Group

Mar 2022 – Sep 2022

Germany

- Developed **FullFormer**, a transformer-based autoregressive implicit generative model that handles both watertight and non-watertight surfaces, enabling generation of nested 3D shapes.
- Published at **DAGM GCPR 2023**; invited talk at SFI Visual Intelligence, Norway.

Student Researcher

University of Siegen

Jan 2021 – Oct 2021

Siegen, Germany

- Applied policy-based reinforcement learning to facility layout planning, minimising material-handling costs.
- Supported an RL system for dynamic flexible job-shop scheduling and built its user interface; developed a web crawler for literature references.

Education

Ph.D. in Machine Learning

University of Mannheim • Advisor: Prof. Dr.-Ing. Margret Keuper

2023 – Early 2027 (expected)

Mannheim, Germany

M.Sc. Mechatronics

University of Siegen

2019 – 2022

Siegen, Germany

B.E. Mechanical Engineering, with Distinction

Chaitanya Bharathi Institute of Technology (CBIT), Osmania University

2014 – 2018

Hyderabad, India

Publications

Full list: Google Scholar

T. Medi, H.-Y. Wang, A. Rampini, M. Keuper. *Decoupling High and Low Frequencies for Faithful Image Generation with Fine Details*. **ECCV 2026**

T. Medi, L. Mikeladze, M. Keuper. *RL-FAT: Reinforcement Learning for Fair Adversarial Training*. **BMVC 2026**

P. Konz, **T. Medi**, M. Keuper. *Unsupervised Anomaly Detection Using Flow Matching on Tabular Data*. MIDAS Workshop, **ECML-PKDD 2026**

T. Medi, A. Rampini, P. Reddy, P. K. Jayaraman, M. Keuper. *3D-WAG: Hierarchical Wavelet-Guided Autoregressive Generation for High-Fidelity 3D Shapes*. **BMVC 2025**

T. Medi, S. Jung, M. Keuper. *FAIR-TAT: Improving Model Fairness Using Targeted Adversarial Training*. **WACV 2025**

T. Medi, S. Jung, M. Keuper. *TRIX: Trading Adversarial Fairness via Mixed Adversarial Training*. arXiv:2507.07768, 2025

T. Medi, J. Tayyub, M. Sarmad, F. Lindseth, M. Keuper. *FullFormer: Generating Shapes Inside Shapes*. **DAGM GCPR 2023**

T. Medi, J. Grabinski, M. Keuper. *Towards Class-wise Robustness Analysis*. OOD Generalization in CV Workshop, **ICCV 2023 (Oral)**

Technical Skills

Programming	Python
Deep Learning	PyTorch, PyTorch Lightning, TensorFlow; transformers, autoregressive models, flow matching, adversarial training
Scale & Infra	Distributed training with Ray, CUDA, multi-GPU HPC clusters
Engineering	Git/GitHub, reproducible research codebases, academic writing
Languages (spoken)	English (fluent)

Talks

- *FullFormer: Generating Shapes Inside Shapes* – Invited talk, SFI Visual Intelligence, Norway, Oct 2023.
- *Towards Class-wise Robustness Analysis* – Oral presentation, ICCV 2023 Workshop, Paris, Oct 2023.

Teaching & Mentoring

- **Thesis supervision:** supervised ~10 Bachelor's and Master's theses in computer vision, generative AI and ML.
- **Teaching assistant:** Generative Computer Vision Models (Fall) and Reinforcement Learning (Spring), University of Mannheim.
- **Seminars:** organised computer vision seminars for Bachelor's and Master's students (2023 – present).

Professional Service

Reviewer: CVPR, ICCV, ECCV, ICLR, WACV, BMVC, TMLR, IEEE TPAMI.

References

Prof. Dr.-Ing. Margret Keuper

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Arianna Rampini

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